Lorenzo Vergata

PhD Student in Civil Engineering | Game Design | Programming | Edge AI

Date of birth:	05/05/2000, Milan, Italy	**Nationality:**	Italian
Email:	lorenzo.vergata@polimi.it	**LinkedIn:**	linkedin.com/in/lorenzo-vergata-b988a5234
GitHub:	github.com/Penzo00	**Game:**	Feed Me, Quack!

Profile

PhD student in Civil Engineering at Politecnico di Milano (since September 2026), working on multi-sensor fusion and self-calibration of MEMS with data assimilation and multi-fidelity techniques. MSc in Engineering Physics with a strong background in materials science, MEMS, data-driven modelling, embedded and edge development, and software prototyping. In parallel, first-year Game Design student at AIV – Accademia Italiana dei Videogiochi, aiming to become a technical designer, with game jam experience as game programmer and game designer.

Experience

PhD Student in Civil Engineering Sep 2026 – present
DICA (Dipartimento di Ingegneria Civile e Ambientale) – Politecnico di Milano, Milan, Italy
Doctoral research on multi-sensor fusion and self-calibration of MEMS and sensors, leveraging data assimilation and multi-fidelity techniques, including deep learning.

Research Intern Apr 2026 – Sep 2026
DICA (Dipartimento di Ingegneria Civile e Ambientale) – Politecnico di Milano, Milan, Italy
Started the research on sensor fusion and self-calibration of MEMS. Concluded on starting the PhD.

Secretary Feb 2023 – Apr 2024
NINE – Nanoscience Innovation and Nanotechnology Engineering (Non-profit), Milan, Italy
Supported administrative tasks and internal communication for a student-led nanoscience association.

Treasurer Mar 2022 – May 2023
NINE – Nanoscience Innovation and Nanotechnology Engineering (Non-profit), Milan, Italy
Managed budget tracking for the association's events and initiatives.

Waiter Mar 2022 – Feb 2024
SC Management SRL (Part-time), Milan, Italy
Provided customer service, order management, and teamwork in a fast-paced hospitality environment.

Education

Politecnico di Milano PhD in Civil Engineering, Sep 2026 – present
DICA – Dipartimento di Ingegneria Civile e Ambientale
Research topics: multi-sensor fusion, self-calibration of MEMS, data assimilation, multi-fidelity modelling.

AIV – Accademia Italiana dei Videogiochi Game Design, 1st year, 2026 – present
Milan, Italy
Goal: technical designer.

Politecnico di Milano MSc in Engineering Physics
Final grade: 107/110
Thesis: MEMS accelerometer calibration via an Unscented Kalman Filter-based identification approach. The work explored an electrical-only alternative to mechanical calibration and reached sensitivity errors below 0.1%.

Politecnico di Milano BSc in Materials Science and Nanotechnology Engineering
Final grade: 102/110

Liceo Alessandro Volta, Milan High school diploma
Final grade: 65/100

Languages

- Italian: native speaker
- English: TOEIC ETS 880/990 (B2), EAS Milan

Certifications & Further Education

- **Grade 3 Music Theory** – ABRSM (June 2013)
  Score: 88/100. Presented for examination by L'ALTRAMUSICA.

Game development

The Weigh of Gold – game jam
Roles: game programmer and game designer. gabris-games.itch.io/the-weigh-of-gold

Severance Package – game jam
Role: game programmer. gaetanostrong.itch.io/severancepackage

Feed Me, Quack! – *POLIMI Game Collective*
A non-violent roguelite released for Windows, macOS, and Linux. I contributed to music and gameplay focused on movement, progression, and resource management. itch.io/feed-me-quack

Selected projects

A-MYCO: Prototyping the Eyes of a Myco-Robot
Chiara D'Amato, Edoardo Torre, Lorenzo Vergata — Politecnico di Milano
Research and prototyping project on mushroom detection and classification for a portable myco-robot. The device uses neural-network inference over mushroom image libraries to support real-time analysis in a compact, low-cost form factor.
Topics: Python, computer vision, neural networks, data preprocessing, edge deployment.

Italian condominium meeting minutes automation
Current project
Python pipeline for an edge device with CPU-only execution and 8 GB RAM. The workflow records from the microphone, stops on Enter, transcribes Italian speech with a fast ASR neural network, rewrites the output with a fast on-device LLM, generates a `.tex` file, compiles it with `pdflatex`, prints the resulting PDF with the default printer through SumatraPDF, and creates a dedicated folder for each meeting so every compilation stays isolated.

Lecture notes

Finished: *Nonlinear Solid Mechanics* (PhD course), *Applied Statistics*, *Machine Learning, Deep Learning and Online Learning*. Work in progress: *La Bibbia del Game Design*, *Advanced Gyroscopes*, *Model Order Reduction Techniques*, *Unreal Engine Fundamentals*. Available on my website.

Open-source repositories

- **AMYCO-Robot** - neural-network-based mushroom detection and classification prototype, with Python and shell utilities for preprocessing, training, and Raspberry Pi deployment.
- **Smith-Waterman** - parallel CUDA implementation of the Smith-Waterman algorithm.
- **pacs-examples** - examples for the Advanced Programming for Scientific Computing (PACS) course at Politecnico di Milano.

Skills

- Python, MATLAB, C#, C++, LaTeX, Bash
- Unreal Engine (C++ and Blueprints), game prototyping
- Git, Linux, Raspberry Pi
- Edge AI and on-device pipelines
- ASR and LLM-based workflows
- Computer vision and data preprocessing
- Scientific programming

Additional activities

- 4-valve trumpet player; performed with **Ibrahim Maalouf** in Bursa and Milan
- Trumpet player at **Banda di Cologno Monzese**
- Member of **Gruppo Micologico Milanese**
- Practiced competitive athletics during high school, with a focus on sprinting and standing jumps

Milan, November 12, 2026

